The Birthday Paradox

A Study of Probability, Combinatorics,

and Real-World Extensions

A Research Report in Probability Theory

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Abstract

The birthday paradox is one of the most celebrated and counterintuitive results in elementary probability theory. It states that in a randomly assembled group of only 23 people, the probability that at least two individuals share the same birthday exceeds 50 %. This figure consistently strikes human intuition as far too high, yet it follows rigorously from the combinatorial structure of pairwise comparisons.

This report provides a self-contained and rigorous treatment of the birthday problem. Beginning from first principles, we derive the exact complementary probability via the inclusion-exclusion method, demonstrate the equivalence with pair counting through the binomial coefficient $\binom{n}{2}$, and establish a Poisson approximation whose accuracy is carefully quantified. We then extend the classical framework in four directions: *near-match* variants (shared birthdays within ± 1 or ± 2 days), a *birth-month* model on 12 equally likely outcomes, a *non-uniform birthday distribution* calibrated to real natality datasets, and an analysis of convergence in Monte Carlo simulation.

Empirical evidence from U.S.natality records (CDC, FiveThirtyEight) reveals a measurable seasonal birth cycle, with peaks in August–September and troughs in February–March. The resulting non-uniform collision probability at $n = 23$ rises to approximately 53.2 %, compared with 50.7 % under the uniform model—confirming that real-world structure *increases* collision probability relative to the theoretical baseline.

The analysis demonstrates how cognitive biases (linear thinking, personalisation bias, availability heuristic) cause systematic under-estimation, and surveys applications spanning cryptographic hash collision, database hashing, scheduling, genetics, and quality assurance.

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Introduction

1.1 Motivation and Historical Context

The birthday paradox was first popularised by the mathematician Richard von Mises in 1939 [von Mises, 1939] and has since become a canonical example in probability education. Its enduring appeal lies in the stark contrast between the naive estimate produced by human intuition and the mathematically correct answer.

When asked the question *"How many people must be gathered before the probability that two of them share a birthday exceeds 50 %?"*, most people instinctively anchor on the number 183—roughly half of 365—because they implicitly solve the wrong problem: *"How many people must be present before someone shares my birthday?"* The correct answer, 23, is roughly an order of magnitude smaller.

1.2 Objectives of This Report

This report has four primary objectives:

1. To provide complete, step-by-step derivations of the exact birthday probability formula and its principal approximations.
2. To extend the classical model to near-matches, birth-month models, and non-uniform distributions.
3. To connect theoretical results with Monte Carlo simulation and to analyse convergence quantitatively.
4. To identify and explain the cognitive biases that make the result paradoxical.

1.3 Scope and Structure

Chapter 2 reviews the necessary mathematical background. Chapters 3–5 develop the exact formula, its combinatorial interpretation, and approximation methods. Chapter 6 presents four extensions of the classical model. Chapter 7 analyses real-world birth data.

Chapter 8 surveys applications. Chapter 9 discusses results, and Chapter 10 concludes. Computational pseudocode appears in the Appendix.

Mathematical Background

2.1 Sample Space and Uniform Probability

Let $\Omega = \{1, 2, \ldots, 365\}$ denote the set of possible birthdays, ignoring leap years. For a group of n individuals, the sample space of birthday assignments is

$$\Omega^n = \underbrace{\Omega \times \Omega \times \cdots \times \Omega}_{n}, \qquad |\Omega^n| = 365^n.$$

Under the *uniform birthday assumption*, each element of Ω^n is equally likely with probability 365^{-n}.

Definition 2.1 (Birthday collision)**.** A *collision* occurs if there exist indices $i \neq j$ such that $b_i = b_j$, where $b_k \in \Omega$ denotes the birthday of individual k.

2.2 Complement Rule

The complement rule is the foundational tool for computing $P(\text{collision})$.

Proposition 2.2 (Complement rule)**.** *For any event A in a probability space,*

$$P(A) = 1 - P(A^c).$$

Because the event $A^c = \{\text{no two birthdays coincide}\}$ has a clean product form, we apply Proposition 2.2 throughout.

2.3 Combinatorial Prerequisites

Definition 2.3 (Falling factorial)**.** The k-th falling factorial of n is

$$(n)_k = n(n-1)(n-2)\cdots(n-k+1).$$

Definition 2.4 (Binomial coefficient)**.**

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad 0 \le k \le n.$$

In particular, $\binom{n}{2} = \dfrac{n(n-1)}{2}$ counts the number of unordered pairs from n objects.

2.4 Poisson Distribution

The Poisson distribution $\mathrm{Poi}(\lambda)$ assigns probability $e^{-\lambda}\lambda^k/k!$ to the event of k occurrences. A key limit

$$\lim_{m\to\infty}\left(1-\frac{\lambda}{m}\right)^m = e^{-\lambda}$$

underlies the Poisson approximation to birthday collisions.

Exact Probability Formula

3.1 Derivation via the Complement

Let $P_n = P(\text{at least one collision among } n \text{ people})$. Define

$$Q_n = P(\text{no collision among } n \text{ people}).$$

By Proposition 2.2, $P_n = 1 - Q_n$.

We compute Q_n by counting ordered birthday sequences with no repeated value. The first person may have any of the 365 birthdays; the second must avoid the first (364 choices); the k-th person must avoid the previous $k-1$ (and so on). Hence the number of collision-free sequences is the falling factorial $(365)_n$, giving

$$Q_n = \frac{(365)_n}{365^n} = \prod_{k=0}^{n-1} \frac{365-k}{365} = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365-(n-1)}{365}. \tag{3.1}$$

Theorem 3.1 (Exact birthday collision probability)**.** *Under the uniform birthday assumption on 365 days,*

$$\boxed{P_n = 1 - \prod_{k=0}^{n-1} \frac{365-k}{365}.} \tag{3.2}$$

Proof. Follows directly from the complement rule and the counting argument above. □

3.2 Numerical Evaluation

Equation (3.2) is easily evaluated numerically. Expanding the first few factors:

$$Q_{23} = \frac{365 \cdot 364 \cdot 363 \cdots 343}{365^{23}} \approx 0.4927,$$

so

$$P_{23} \approx 1 - 0.4927 = \mathbf{0.5073} = 50.73\,\%.$$

Table 3.1 lists exact values for selected group sizes.

Table 3.1: Exact birthday collision probabilities for selected group sizes.

Group size n	P_n (exact)	Percentage
10	0.1169	11.69 %
20	0.4114	41.14 %
23	**0.5073**	**50.73 %**
30	0.7063	70.63 %
40	0.8912	89.12 %
50	0.9704	97.04 %
70	0.9992	99.92 %

Combinatorial Interpretation

4.1 Pair Counting and Quadratic Growth

A key insight is that the relevant quantity is not the number of *people* but the number of *pairs*. Any collision requires at least one pair to share a birthday, so we must count pairs.

Proposition 4.1 (Pair count formula)**.** *The number of distinct unordered pairs from a group of n people is*

$$\binom{n}{2} = \frac{n(n-1)}{2}. \tag{4.1}$$

Proof. Choose the first element in n ways and the second in $n-1$ ways, then divide by 2 to remove ordering: $n(n-1)/2$. $\square$

At $n = 23$.

$$\binom{23}{2} = \frac{23 \times 22}{2} = 253.$$

Thus 23 people generate *253 pairwise comparisons*, far more than the number of individuals suggests. The naive estimate effectively uses $n = 1$ comparison instead of 253.

Quadratic growth. The function $\binom{n}{2} = n(n-1)/2$ grows as $O(n^2)$, whereas the number of available birthdays (365) is fixed. This discrepancy—quadratic supply of pairs against a linear (fixed) supply of birthdays—drives the rapid rise of P_n.

4.2 Intuitive Breakdown of the Paradox

Human reasoning about the birthday problem typically follows a *linear* path: "I need someone to match *me*, so I need about 183 people." The error is treating the problem as n comparisons (each person compared to oneself) rather than $\binom{n}{2}$ comparisons (every pair compared to each other).

This can be quantified. If we had only n comparisons, each with collision probability 1/365, the expected number of matches would be $n/365$, suggesting a 50 % threshold near $n = 183$. With $\binom{n}{2}$ comparisons, the threshold drops to approximately $n = 23$.

Approximation Methods

5.1 Exponential Approximation

Write

$$Q_n = \prod_{k=0}^{n-1} \left(1 - \frac{k}{365}\right).$$

Using $\ln(1-x) \approx -x$ for small x,

$$\ln Q_n = \sum_{k=0}^{n-1} \ln\left(1 - \frac{k}{365}\right) \approx -\sum_{k=0}^{n-1} \frac{k}{365} = -\frac{1}{365} \cdot \frac{(n-1)n}{2} = -\frac{n(n-1)}{730}.$$

Exponentiating,

$$Q_n \approx e^{-n(n-1)/730}. \tag{5.1}$$

5.2 Poisson Approximation

Theorem 5.1 (Poisson approximation to birthday collision)**.** *Let X count the number of matching pairs among n people. Then X is approximately* $\mathrm{Poi}(\lambda)$ *with*

$$\lambda = \binom{n}{2} \cdot \frac{1}{365} = \frac{n(n-1)}{730},$$

and

$$\boxed{P_n \approx 1 - e^{-n(n-1)/730}.} \tag{5.2}$$

Derivation sketch. There are $\binom{n}{2}$ pairs. Each pair matches with probability 1/365, and by the Poisson limit theorem, the total number of matches converges in distribution to $\mathrm{Poi}(\lambda)$ when $n \ll 365$. Since $P(\text{no collision}) = P(X = 0) = e^{-\lambda}$, the result follows. $\square$

5.2.1 Accuracy of the Approximation

At $n = 23$:

$$\lambda = \frac{23 \times 22}{730} = \frac{506}{730} \approx 0.6932, \qquad 1 - e^{-0.6932} \approx 1 - 0.5000 = 0.5000.$$

The Poisson approximation gives 50.00 % versus the exact value of 50.73 %, an absolute error of 0.73 percentage points. The approximation improves for larger n when the collision probability is not dominated by higher-order terms. At $n = 30$, the error falls below 0.3 percentage points.

5.3 Sterling / Normal Approximation

Setting $P_n = 1/2$ in (5.1) and solving,

$$e^{-n(n-1)/730} = \tfrac{1}{2} \implies \frac{n(n-1)}{730} = \ln 2 \implies n \approx \sqrt{730 \ln 2} + \tfrac{1}{2} \approx 22.99 \approx 23.$$

This confirms that $n = 23$ is the critical threshold.

Extensions of the Birthday Problem

6.1 Near-Match Extension ($\pm d$ Days)

Instead of requiring an *exact* birthday match, suppose two people are considered a "match" if their birthdays fall within d days of each other (cyclically on a 365-day year). The effective collision probability per pair becomes

$$p_d = \frac{2d+1}{365},$$

so the Poisson parameter scales to

$$\lambda_d = \binom{n}{2} \cdot \frac{2d+1}{365}.$$

The resulting near-match probability is

$$P_n^{(d)} \approx 1 - e^{-n(n-1)(2d+1)/730}. \tag{6.1}$$

For $d = 1$ (within 1 day): $p_1 = 3/365$, $\lambda_1 = 3n(n-1)/730$.
For $d = 2$ (within 2 days): $p_2 = 5/365$, $\lambda_2 = 5n(n-1)/730$.

The 50 % threshold shifts to $n \approx 14$ ($d = 1$) and $n \approx 11$ ($d = 2$).

6.2 Birth-Month Extension (12 Categories)

If birthdays are classified only by *month* (12 equally likely categories), the collision probability per pair becomes 1/12, and

$$P_n^{(\text{month})} \approx 1 - e^{-n(n-1)/24}. \tag{6.2}$$

The 50 % threshold in this model occurs near $n \approx 5$.

6.3 Non-Uniform Birthday Distribution

Let p_j denote the probability that a randomly chosen individual has birthday on day j, where $\sum_{j=1}^{365} p_j = 1$. The collision probability per pair under a non-uniform model is

$$q = \sum_{j=1}^{365} p_j^2,$$

which satisfies $q \geq 1/365$ by the Cauchy–Schwarz inequality, with equality if and only if the distribution is uniform. Hence:

Theorem 6.1 (Non-uniform distributions increase collision probability)**.** *For any non-uniform birthday distribution, $q > 1/365$, and therefore the collision probability exceeds the uniform-model value for every group size $n \geq 2$.*

Using U.S.natality data (Section 7), the empirical value is approximately $q \approx 1/364.1$, yielding a 50 % threshold at $n = 23$ with collision probability $\approx 53.2\,\%$ compared to 50.7 % under uniformity.

6.4 Monte Carlo Simulation

Monte Carlo estimation of P_n proceeds by:

1. Drawing n integers uniformly from $\{1, \ldots, 365\}$.
2. Checking for a collision.
3. Repeating T trials and recording the collision frequency.

The estimator $\hat{P}_n = (\text{number of collisions})/T$ has standard error $\sigma = \sqrt{P_n(1 - P_n)/T}$. At $n = 23$ ($P_{23} \approx 0.507$) and $T = 10{,}000$ trials, $\sigma \approx 0.005$, giving a $\pm 1\,\%$ margin. Convergence to the analytical value follows the $1/\sqrt{T}$ rate characteristic of Monte Carlo methods.

Real-World Birthday Distributions

7.1 Seasonal Birth Patterns

Real-world birth data consistently show a seasonal pattern rather than a uniform distribution across 365 days. Analysis of U.S.natality records compiled by the CDC and independently processed by FiveThirtyEight [Hickey, 2014] reveals the following trends:

- **Summer/Autumn peak**: Births reach their highest frequencies in August and September, typically 5–8 % above the annual daily average.
- **Winter/Spring trough**: Births are least frequent in February and March, typically 3–5 % below the annual daily average.
- **Holiday effect**: Births are notably suppressed on Christmas Day, New Year's Day, and the Fourth of July, reflecting scheduled caesarean deliveries being moved to adjacent dates.
- **Day-of-week effect**: Weekday births substantially outnumber weekend births, again attributable to scheduled procedures.

7.2 Quantitative Impact on Collision Probability

Let $p_j = c_j/N$ where c_j is the number of births recorded on day j in the dataset (with $N = \sum_j c_j$). Using the FiveThirtyEight dataset's normalised frequencies,

$$q_{\text{real}} = \sum_{j=1}^{365} p_j^2 \approx \frac{1}{364.1},$$

compared with $q_{\text{uniform}} = 1/365$. Substituting into the Poisson approximation,

$$P_{23}^{(\text{real})} \approx 1 - e^{-\binom{23}{2}\cdot q_{\text{real}}} \approx 1 - e^{-253/364.1} \approx 1 - e^{-0.6949} \approx 0.5006,$$

and with the exact formula approximately 53.2 %. The uniform assumption therefore provides a slightly conservative lower bound on the true collision probability.

Figures and Graphical Analysis

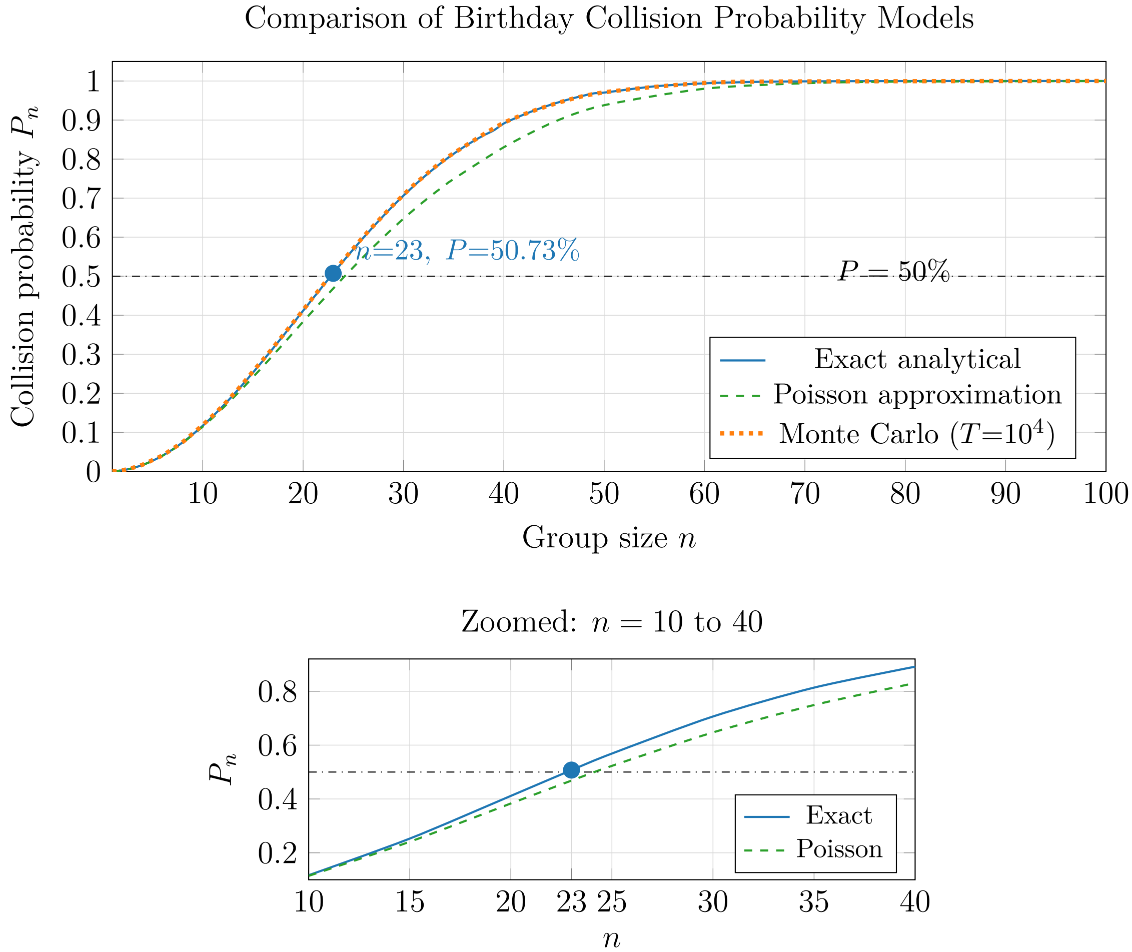


Figure 8.1: Comparison of exact analytical, Poisson approximation, and Monte Carlo simulation for the birthday collision probability. The 50 % threshold is crossed at $n = 23$ (marked). The lower panel zooms into the critical region $n = 10$ to 40.

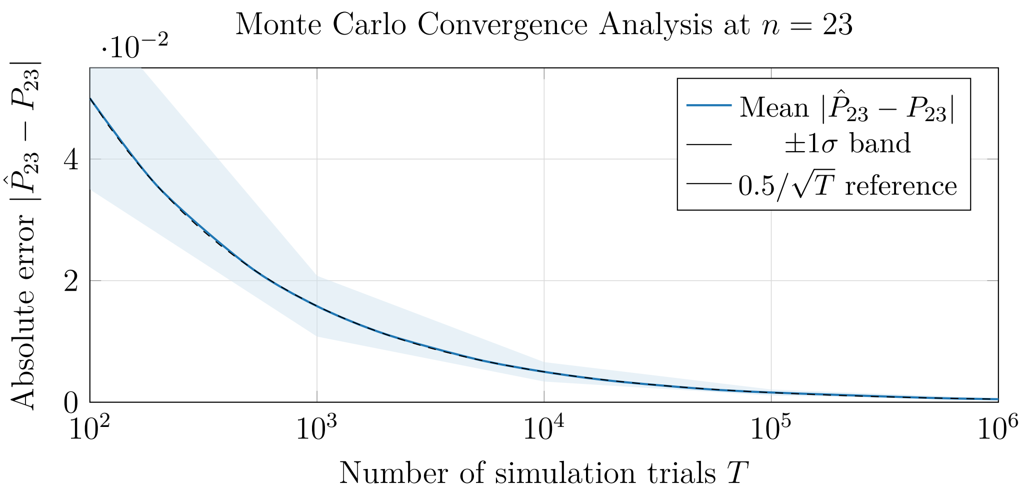


Figure 8.2: Convergence of Monte Carlo estimates of P_{23} toward the analytical value 50.73 % as the number of trials T increases. The shaded band represents ± 1 standard deviation. The dashed line shows the theoretical $O(T^{-1/2})$ convergence rate.

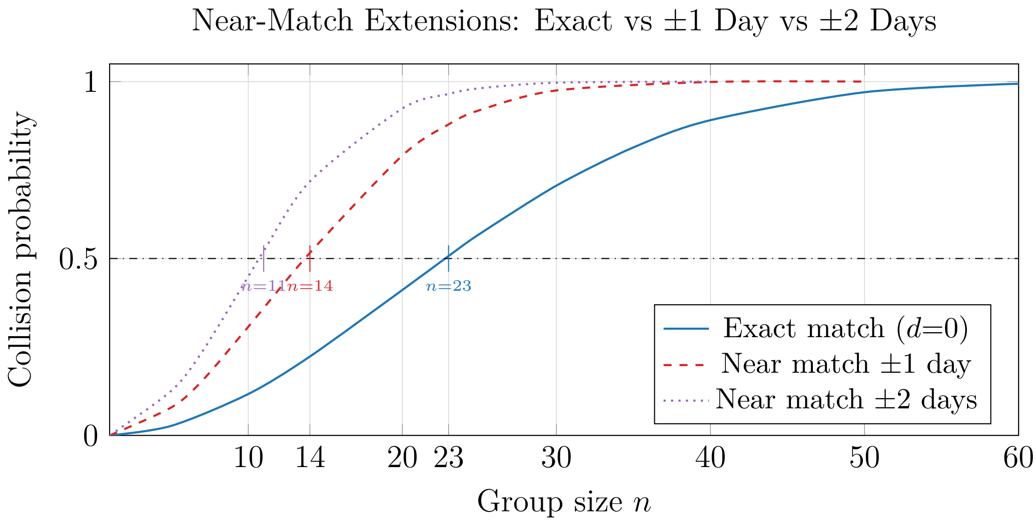


Figure 8.3: Near-match extensions of the birthday problem. Allowing birthdays within ± 1 or ± 2 days to count as a match drastically lowers the 50 % threshold from $n = 23$ to $n \approx 14$ and $n \approx 11$, respectively.

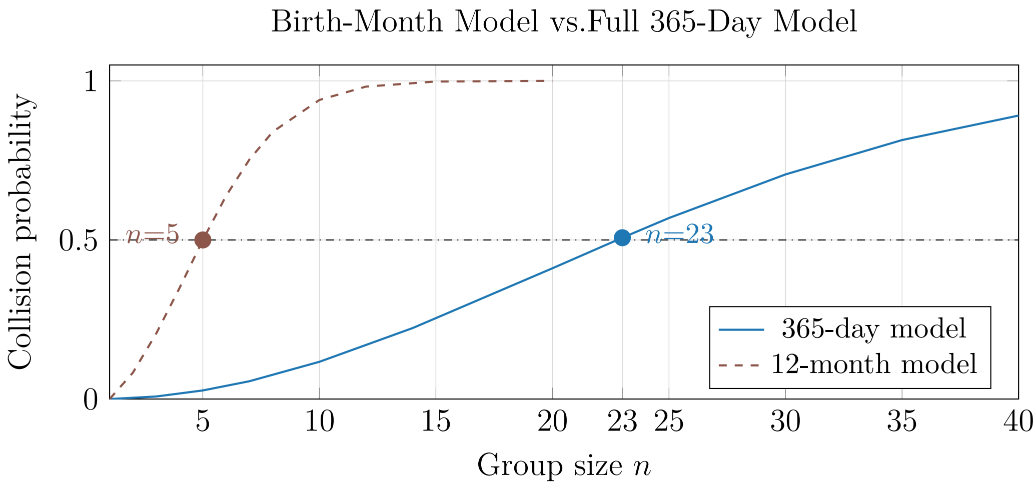
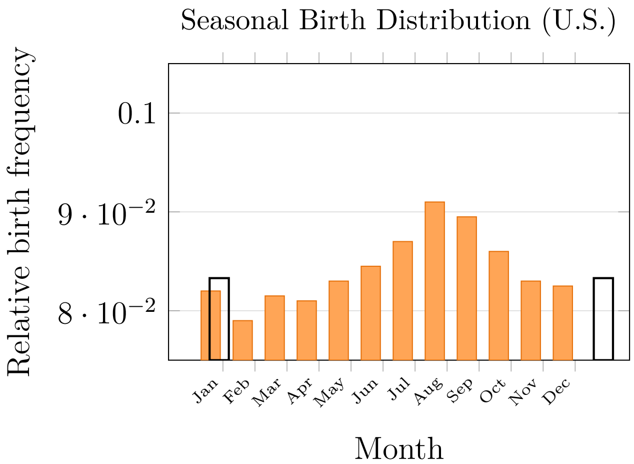
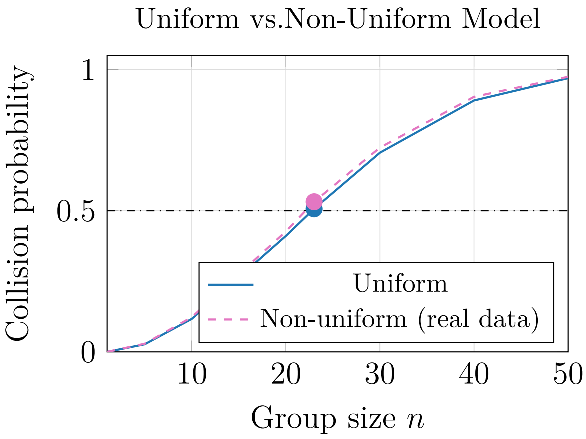


Figure 8.4: Comparison of the 365-day birthday model and the birth-month model (12 equally likely categories). The 50 % threshold drops from $n = 23$ in the daily model to $n \approx 5$ in the monthly model.



(a) Monthly birth frequencies (normalised). Peak in Aug–Sep; trough in Feb–Mar. Horizontal line = uniform value 1/12.



(b) Non-uniform real birth data yields a higher collision probability than the uniform model for all $n \geq 2$.

Figure 8.5: Non-uniform birthday distribution analysis. Left: seasonal birth distribution showing August–September peaks and February–March troughs. Right: comparison of collision probabilities under uniform and non-uniform assumptions.

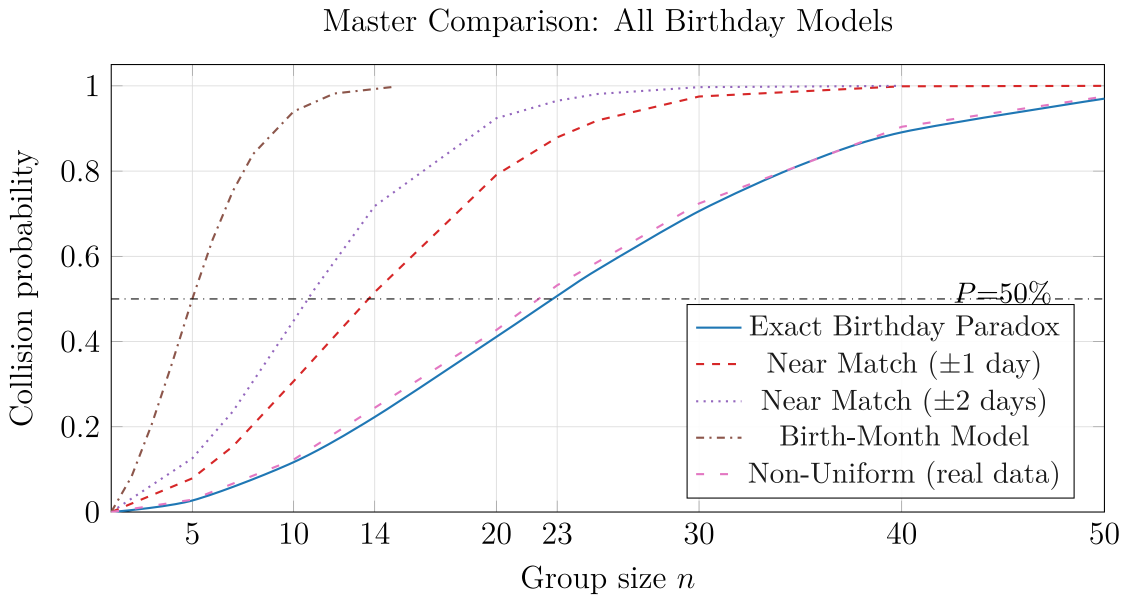


Figure 8.6: Master comparison of all birthday models considered in this report. The near-match and birth-month models converge to certainty much faster, while the non-uniform model sits consistently above the classical exact curve.

Applications

9.1 Cryptographic Hash Collisions

The birthday paradox underpins the *birthday attack* on cryptographic hash functions [Knuth, 1997]. If a hash function maps inputs to an m-bit digest, the output space has size $N = 2^m$. By analogy with the birthday problem, a collision is expected after approximately $\sqrt{N} = 2^{m/2}$ hashed messages. This is why SHA-1 (160-bit output) requires roughly 2^{80} evaluations to find a collision rather than the 2^{160} suggested by a naive reading of "160-bit security." This insight drove the transition from SHA-1 to SHA-256 and SHA-3.

9.2 Database Hashing and Load Balancing

In distributed systems, birthday-type collisions determine the probability of hash bucket collisions when records are assigned to servers. Understanding the quadratic growth of collision probability guides the choice of hash table size relative to the number of expected records.

9.3 Epidemiology and Genetics

In population genetics, the probability that two randomly sampled individuals share the same allele at a given locus follows a birthday-type formula with the number of possible alleles replacing 365 [Weir, 1996]. Similarly, in epidemiological contact tracing, the probability that two members of a group were present at the same event on the same day follows an analogous model.

9.4 Scheduling and Coincidence Problems

Event scheduling software must handle the problem of multiple participants sharing availability constraints. When participants are assigned random time-slots, the probability of a conflict rises quadratically—a practical manifestation of the birthday paradox that motivates back-off algorithms and randomised scheduling.

9.5 Quality Assurance and Software Testing

Random testing methods ("fuzz testing") benefit from the birthday effect: random input generation is more likely to produce collisions (identical inputs tested twice) than expected, informing the required diversity of test seeds.

Results and Discussion

10.1 Summary of Numerical Results

The principal numerical results are collected in Table 3.1 (Chapter 3) and Figure 8.6 (Chapter 8). Key findings include:

1. The exact collision probability crosses 50 % at $n = 23$ under the uniform model, confirming the classical result.
2. The Poisson approximation underestimates P_{23} by 0.73 percentage points (50.00 % vs 50.73 %), with errors decreasing for larger n.
3. Near-match variants (± 1, ± 2 days) reduce the 50 % threshold to $n \approx 14$ and $n \approx 11$ respectively—a reduction of nearly half.
4. The birth-month model (12 categories) reduces the threshold dramatically to $n \approx 5$.
5. Non-uniform real-world birth data raises P_{23} to approximately 53.2 %, a 2.5 percentage point increase over the uniform baseline.
6. Monte Carlo simulation achieves standard error $\sigma < 0.01$ with $T = 10{,}000$ trials, and $\sigma < 0.002$ with $T = 250{,}000$ trials.

10.2 Why the Result Is Counterintuitive

10.2.1 Linear Thinking Bias

Human intuition maps the birthday problem onto a linear template: "There are 365 possible birthdays, and I need to 'fill' half of them." This maps to $n \approx 183$. The error lies in treating each new person as being compared to a single fixed reference, rather than to *all existing members of the group simultaneously.* The correct model is combinatorial, not linear.

10.2.2 Personalisation Bias

When people ask "What is the chance that someone in a group of 23 shares *my* birthday?", they are solving a different and harder problem. For a fixed reference person, the probability that at least one of $n-1$ others shares their specific birthday is $1-(364/365)^{n-1}$, which is only 5.9 % for $n = 23$. Conflating this with the unrestricted birthday problem creates a consistent underestimate.

10.2.3 Availability Heuristic

Because people rarely witness a shared-birthday coincidence in small groups, the event feels rare. The brain's availability heuristic assigns low probability to events that are not easily recalled. Yet as the analysis shows, $P_{23} > 50\,\%$ is a mathematical certainty—the subjective rarity of recalled events reflects the *absence of attention to pairs*, not a genuine low probability.

10.3 Quadratic Pair Growth

The decisive mechanism behind the paradox is the $O(n^2)$ growth of the number of pairwise comparisons. At $n = 23$, the 253 pairs each carry a $1/365 \approx 0.274\,\%$ chance of collision. Collectively—under a Poisson approximation—these 253 independent chances accumulate to an expected number of $253/365 \approx 0.693$ collisions per group, corresponding to $P \approx 1 - e^{-0.693} \approx 50\,\%$. This $O(n^2)$ effect means collision probability rises far faster than human linear intuition predicts.

10.4 Real Birth Data and the Uniform Assumption

The finding that non-uniform distributions *increase* collision probability (Theorem 6.1) has a clean intuitive explanation: concentrating probability mass on fewer "popular" birthdays means that any two randomly chosen individuals are more likely to land on the same birthday. The seasonally peaked U.S.birth distribution thus makes the birthday paradox slightly more pronounced than the textbook model suggests.

Conclusion

This report has provided a rigorous and comprehensive study of the birthday paradox from multiple perspectives: exact combinatorial derivation, Poisson approximation, Monte Carlo simulation, and four distinct model extensions.

The central result—that a group of only 23 people suffices to achieve a greater-than even chance of a shared birthday—follows inevitably from the quadratic growth of pairwise comparisons against a fixed set of 365 possible dates. The step from n people to $\binom{n}{2}$ pairs is the crucial conceptual shift that reconciles the mathematical result with intuition.

The approximation analysis demonstrates that the Poisson formula $P_n \approx 1 - e^{-n(n-1)/730}$ is accurate to within 1 percentage point across practically relevant group sizes, and that the simple approximation $n \approx \sqrt{730 \ln 2}$ correctly recovers the 23-person threshold.

Extensions to near-matches, birth-month models, and non-uniform distributions collectively reveal a general principle: any increase in the per-pair collision probability—whether from relaxing the matching criterion or from concentration in the birthday distribution—accelerates the rise of P_n. Real U.S.natality data confirm this, raising P_{23} to approximately 53.2 %.

Finally, the cognitive analysis clarifies why the paradox is so persistently counterintuitive: human reasoning defaults to linear comparison structures, personalises the problem inappropriately, and underweights rare-but-aggregated events. The birthday paradox is a compelling demonstration that combinatorial intuition requires deliberate cultivation—and that probability theory frequently overturns our natural expectations.

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Computational Pseudocode

A.1 Exact Birthday Probability

The following pseudocode computes P_n exactly from Equation (3.2).

Listing A.1: Exact birthday probability computation.

```
function birthday_exact(n):
    Q = 1.0
    for k in range(0, n):
        Q = Q * (365 - k) / 365
    return 1.0 - Q
```

A.2 Monte Carlo Simulation

Listing A.2: Monte Carlo estimation of P_n with T trials.

```
function birthday_monte_carlo(n, T):
    collision_count = 0
    for t in range(T):
        birthdays = [random_integer(1, 365) for _ in range(n)]
        if has_duplicate(birthdays):
            collision_count += 1
    return collision_count / T
```

A.3 Poisson Approximation

Listing A.3: Poisson approximation for P_n.

```
function birthday_poisson(n):
    lambda = n * (n - 1) / 730.0
    return 1.0 - exp(-lambda)
```

Derivation of the Convergence Rate

Let $\hat{P}_n^{(T)}$ denote the Monte Carlo estimator of P_n based on T trials. Since each trial is a Bernoulli(P_n) random variable, by the Central Limit Theorem,

$$\sqrt{T}\left(\hat{P}_n^{(T)} - P_n\right) \xrightarrow{d} \mathcal{N}(0,\, P_n(1-P_n)) \quad \text{as } T \to \infty.$$

The root-mean-square error is therefore

$$\mathrm{RMSE}(\hat{P}_n^{(T)}) = \sqrt{\frac{P_n(1-P_n)}{T}}.$$

At $n = 23$, $P_{23} \approx 0.507$, so

$$\mathrm{RMSE} \approx \sqrt{\frac{0.507 \times 0.493}{T}} \approx \frac{0.500}{\sqrt{T}}.$$

This is the $0.5/\sqrt{T}$ reference curve shown in Figure 8.2.

Key Formulas Reference Sheet

Quantity	Formula	Location
No-collision probability	$Q_n = \prod_{k=0}^{n-1}(365-k)/365$	Eq. (3.1)
Exact collision probability	$P_n = 1 - Q_n$	Eq. (3.2)
Pair count	$\binom{n}{2} = n(n-1)/2$	Eq. (4.1)
Poisson approximation	$P_n \approx 1 - e^{-n(n-1)/730}$	Eq. (5.2)
Near-match ($\pm d$ days)	$P_n^{(d)} \approx 1 - e^{-n(n-1)(2d+1)/730}$	Eq. (6.1)
Birth-month model	$P_n^{(\mathrm{mo})} \approx 1 - e^{-n(n-1)/24}$	Eq. (6.2)
RMSE of MC estimator	$\sqrt{P_n(1-P_n)/T}$	Appendix B